PROJECT REPORT: HR ATTRITION PREDICTION & ANALYTICS

Project Title: Predictive Modeling for Employee Retention

Domain: Data Analytics

Tools & Technologies: Python (Pandas, Scikit-Learn), SQL, Power BI, Excel

1. ABSTRACT

In the modern corporate landscape, human capital is the most significant asset of any organization. However, high employee attrition rates—the phenomenon where employees leave the workforce—pose a massive financial and operational burden. This project aims to move from a reactive "exit interview" approach to a proactive "predictive analytics" model. By leveraging historical employee data, including demographics, satisfaction scores, and performance metrics, we developed a machine learning framework to identify at-risk employees. The project concludes with a Power BI dashboard designed to provide HR managers with real-time, actionable insights into workforce stability.

2. INTRODUCTION

Employee attrition is more than just a HR metric; it is a business crisis. Replacing a mid-level employee can cost an organization up to 150% of that employee's annual salary when considering recruitment, onboarding, and lost productivity.

The objective of this project is to build a predictive system that can:

1. Identify the primary drivers of attrition (e.g., Overtime, Salary, Work-Life Balance).
2. Classify current employees into "At-Risk" or "Stable" categories.
3. Visualize these trends for departmental leadership to implement targeted retention strategies.

3. PROBLEM STATEMENT

Despite having access to vast amounts of employee data, many HR departments struggle to interpret the underlying patterns that lead to resignations. Decisions are often based on intuition rather than data. There is a critical need for a systematic approach to analyze employee behavior and predict future turnover to reduce costs and maintain organizational morale.

4. METHODOLOGY

The project followed the **CRISP-DM** (Cross-Industry Standard Process for Data Mining) framework:

4.1 Data Acquisition

The primary dataset used was the **HR Analytics Employee Attrition & Performance** dataset, consisting of 1,470 employee records and 35 features. Key variables included Age, Business Travel, Daily Rate, Department, Distance From Home, Education, Environment Satisfaction,

Gender, Job Involvement, Job Role, Job Satisfaction, Marital Status, Monthly Income, and Overtime.

4.2 Data Preprocessing (Python)

Raw data is rarely ready for modeling. The following steps were taken using Python (Pandas & NumPy):

- **Handling Missing Values:** Checked for null entries to ensure data integrity.
- **Feature Encoding:** Categorical variables (e.g., "Sales", "Research & Development") were converted into numerical formats using Label Encoding and One-Hot Encoding.
- **Feature Scaling:** Continuous variables like "Monthly Income" were scaled to ensure the model wasn't biased toward larger numerical values.

4.3 Exploratory Data Analysis (EDA)

EDA was performed to find hidden correlations. Key insights included:

- **Overtime:** A clear correlation was found between frequent overtime and high attrition.
- **Income Slabs:** Attrition was highest in the lower-income brackets (under $3,000/month).
- **Age Factor:** Younger employees (ages 18–25) showed a higher tendency to leave compared to senior staff.

5. MODEL DEVELOPMENT & IMPLEMENTATION

The core of the project involved building a classification model.

5.1 Algorithm Selection

We tested multiple algorithms to find the highest accuracy and recall:

- **Logistic Regression:** Served as the baseline model for binary classification.
- **Random Forest Classifier:** Used to handle non-linear relationships and provide "Feature Importance" scores.
- **Decision Trees:** Helpful for visualizing the decision-making path of an employee leaving.

5.2 Performance Metrics

The models were evaluated based on:

- **Accuracy:** The percentage of correct predictions.
- **Recall:** Crucial for this project, as we want to capture as many "At-Risk" employees as possible, even at the cost of some false positives.

- **F1-Score:** The balance between precision and recall.

6. DATA VISUALIZATION (POWER BI)

To make the data accessible to non-technical HR managers, a Power BI dashboard was developed.

Dashboard Storyboard:

- **KPI Cards:** Total Employees, Active Employees, Attrition Count, and Average Age.
- **Attrition by Department:** A bar chart showing that the Sales department typically has the highest turnover.
- **Salary Slab Analysis:** A donut chart showing attrition across different income levels.
- **Risk Heatmap:** A matrix visual crossing "Job Satisfaction" with "Years at Company" to identify hotspots of dissatisfaction.

7. RESULTS & DISCUSSION

The final model achieved an accuracy of approximately **85%**. The most significant predictors of attrition identified were:

1. **Overtime (Highest Impact):** Employees working extra hours consistently are the most likely to exit.
2. **Job Level:** Entry-level roles have significantly higher turnover than managerial roles.
3. **Monthly Income:** Competitive pay remains a dominant factor in retention.
4. **Stock Option Level:** Employees with no stock options were more likely to leave than those with a stake in the company.

8. RECOMMENDATIONS & RETENTION STRATEGIES

Based on the data, the following strategies are proposed:

- **Overtime Mitigation:** HR should audit departments with high overtime and consider hiring seasonal staff or improving workflow efficiency.
- **Career Pathing:** Since "Years in Current Role" was a factor, the organization should implement a "18-month Career Review" to discuss growth opportunities.
- **Targeted Incentives:** Financial incentives or stock options should be prioritized for high-performing employees in the "At-Risk" salary brackets.

9. CONCLUSION

This project demonstrates that employee attrition is not a random occurrence but a predictable outcome influenced by specific organizational factors. By using Python for predictive modeling and Power BI for visualization, HR can transition from guessing why employees leave to knowing who is likely to leave next. This proactive stance saves the company significant costs and fosters a healthier, more stable work environment.

10. FUTURE SCOPE

Future versions of this project could incorporate **Sentiment Analysis** from internal communication tools (like Slack or Teams) to gauge employee mood in real-time. Additionally, integrating a **real-time API** from the company's Payroll and Attendance system would allow the dashboard to update automatically every month.